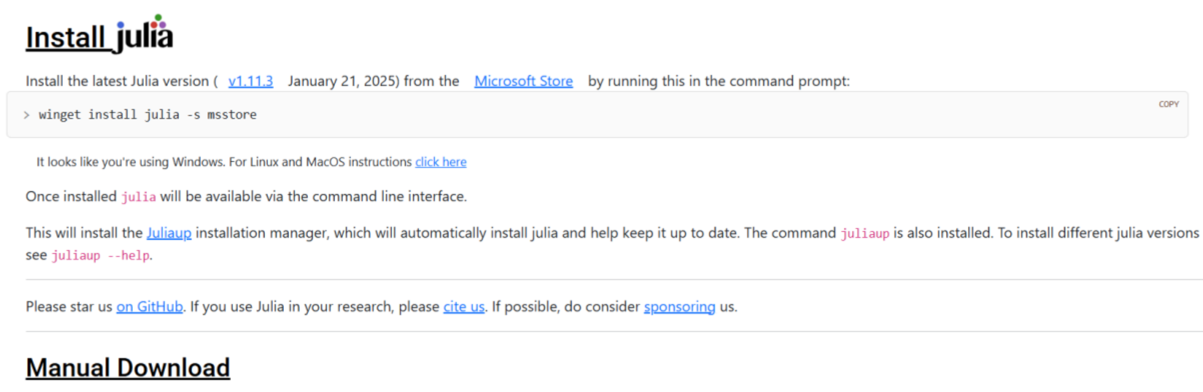
Replication Instructions

Below, we provide detailed instructions for installing the necessary software and running the replication files to reproduce the paper's results on a PC with Microsoft Windows. A short overview of how to install software under macOS is provided at the end of the document. All replication has been tested in Windows and macOS running Julia 1.11, Matlab R2024b, and Dynare 5.5 and 6.3.

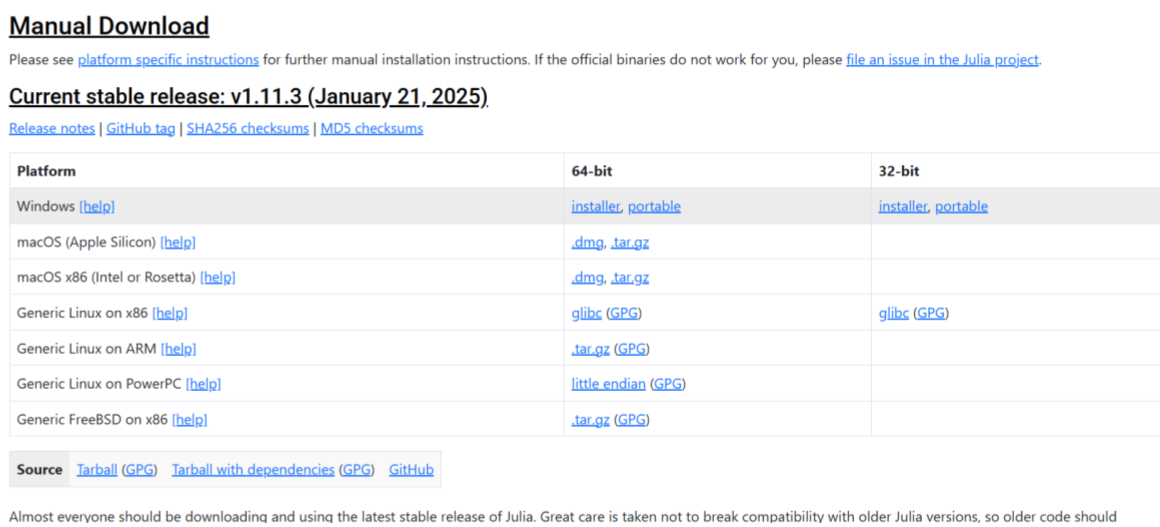
Microsoft Windows

Installing Julia

1. Go to https://julialang.org/downloads/ which provides the Julia installation instructions. For Windows, as of February 27 2025, the website shows the following instructions.



2. We recommend to follow the Manual Download way of installing Julia. Scroll down to the header “Manual Download”. This will provide a table of the most recent Julia version for various operating systems. The table should looks something like this



Manual Download

Please see platform specific instructions for further manual installation instructions. If the official binaries do not work for you, please file an issue in the Julia project.

Current stable release: v1.11.3 (January 21, 2025)

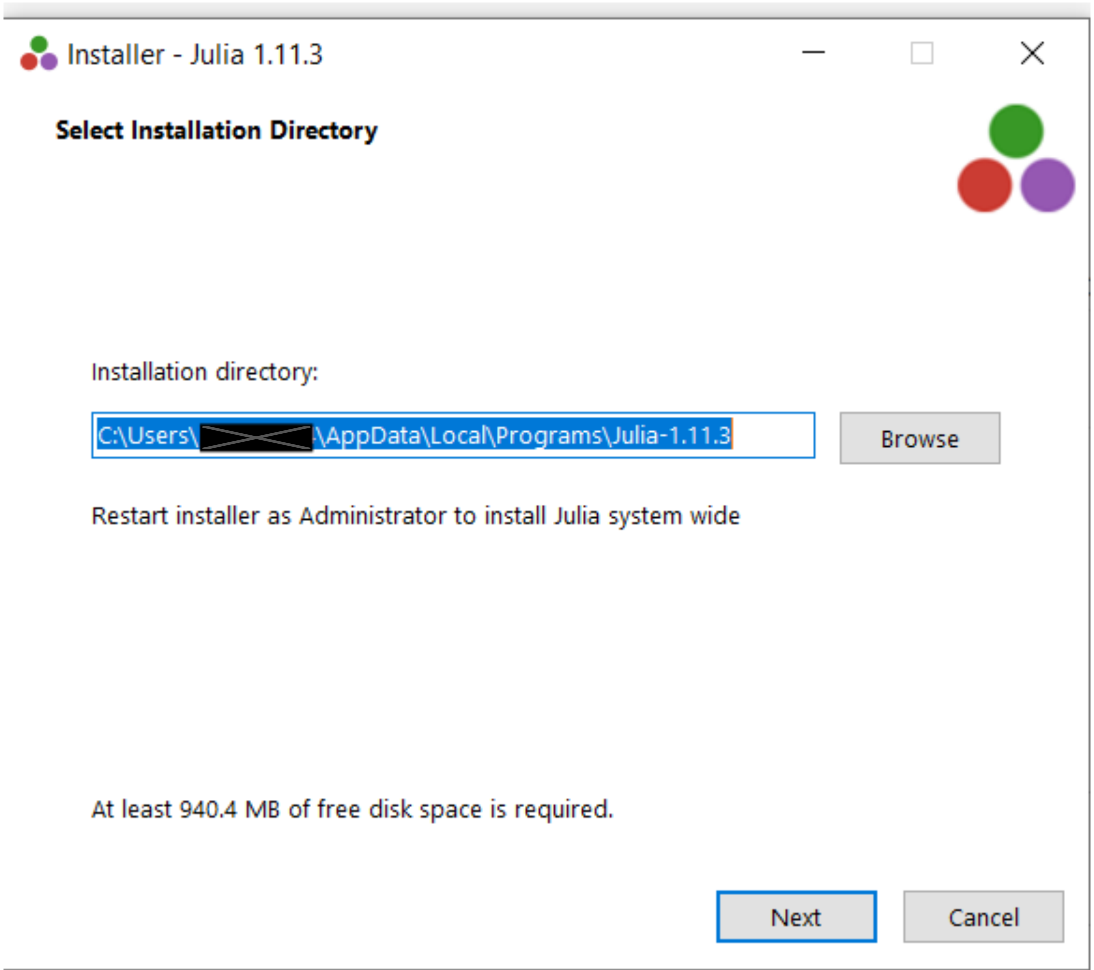
Release notes | GitHub tag | SHA256 checksums | MD5 checksums

Platform	64-bit	32-bit
Windows [help]	installer, portable	installer, portable
macOS (Apple Silicon) [help]	.dmg, .tar.gz	
macOS x86 (Intel or Rosetta) [help]	.dmg, .tar.gz	
Generic Linux on x86 [help]	glibc (GPG)	glibc (GPG)
Generic Linux on ARM [help]	.tar.gz (GPG)	
Generic Linux on PowerPC [help]	little endian (GPG)	
Generic FreeBSD on x86 [help]	.tar.gz (GPG)	

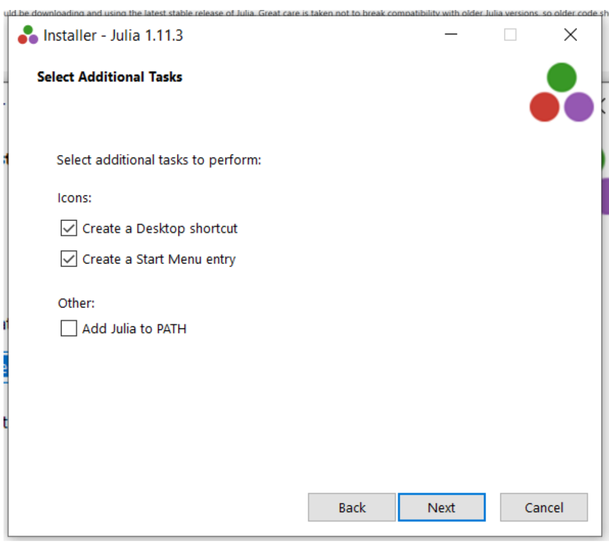
Source Tarball (GPG) Tarball with dependencies (GPG) GitHub

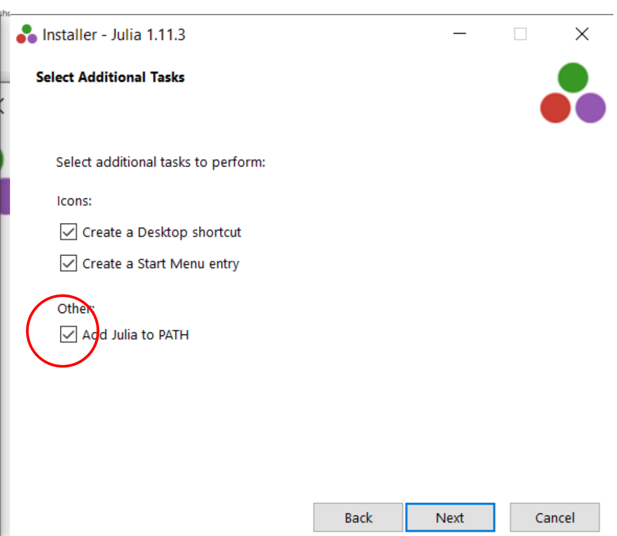
Almost everyone should be downloading and using the latest stable release of Julia. Great care is taken not to break compatibility with older Julia versions, so older code should

3. To download Julia for Windows, click on the “installer” link in the second column “64-bit” or the “installer” link under the third column “32-bit” depending on which system you have. Most modern systems are 64-bit. This will download a Julia installer which should now appear in your Download folder.
4. Double click the installer in the Download folder to open the Julia Installer. This should give you something like the following. Note, your exact version may be different.

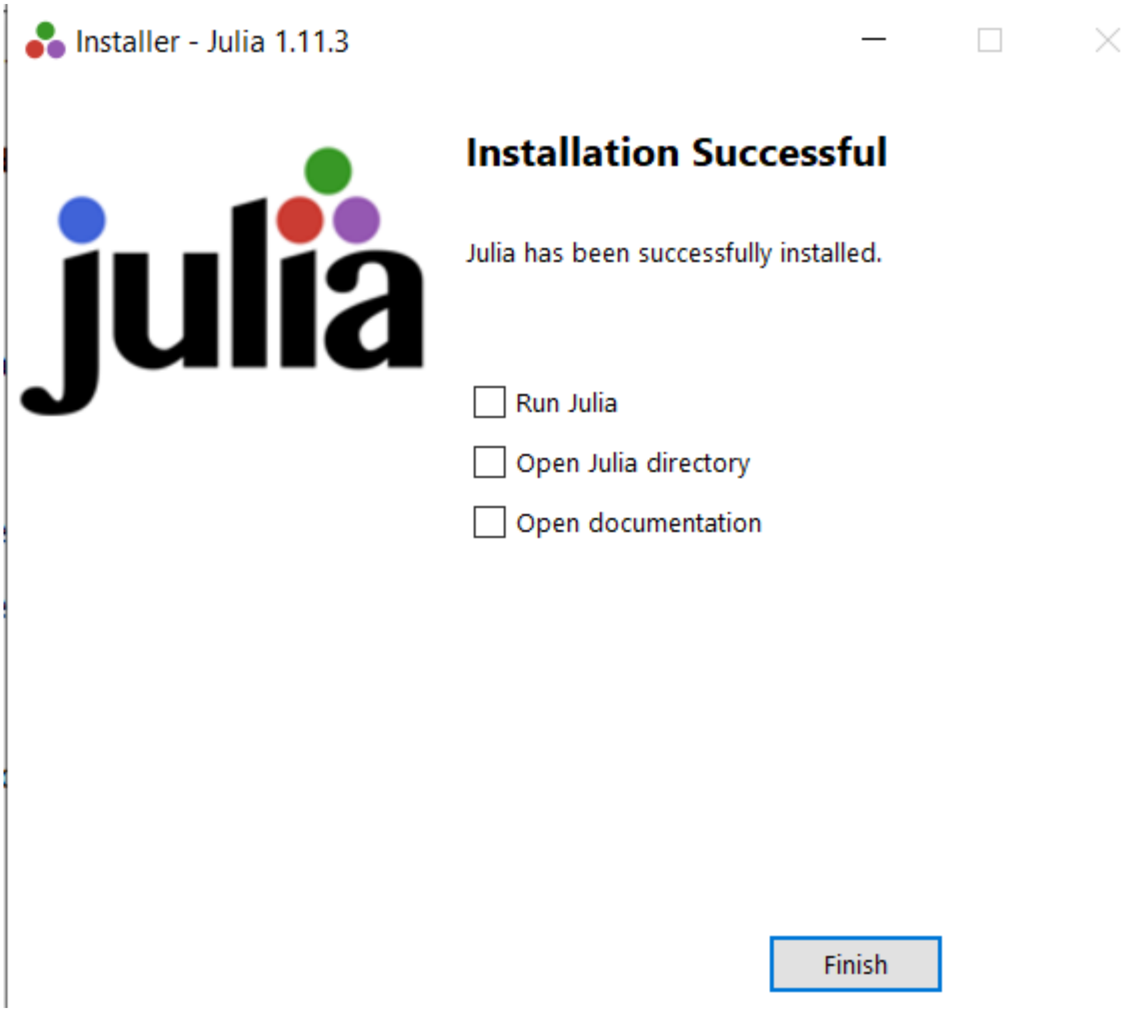


5. The default location is fine for most people. If you want to change the installation location, press "Browse" and choose a different location.
6. Press Next to arrive at the following screen. Select "Add Julia to PATH".

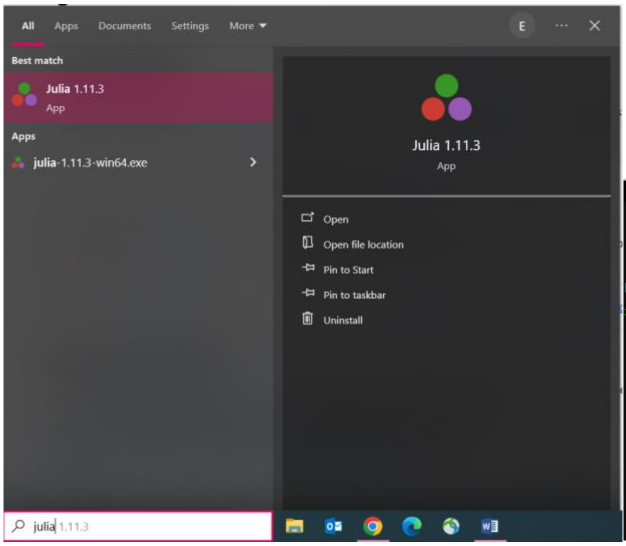


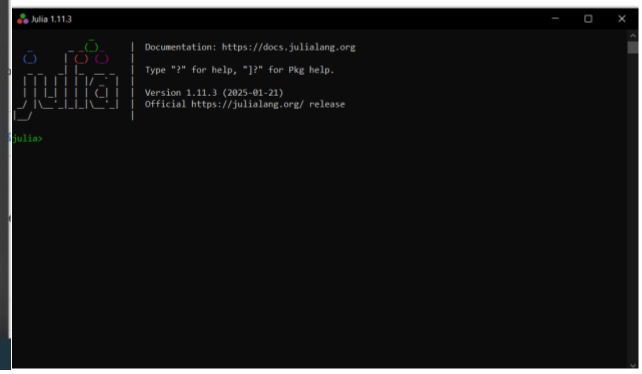


7. Press Next to install Julia. Upon finishing the installation, Julia will present you the following screen. You can simply press Finish.

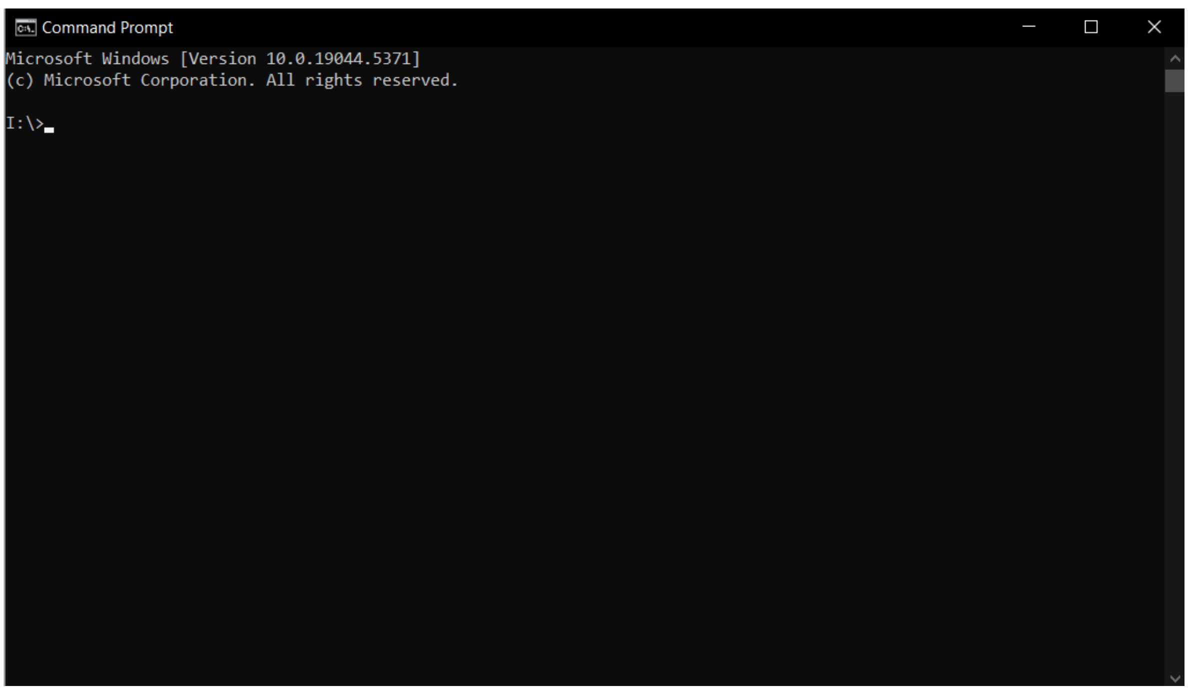


8. To check if the Julia installation was successful, press the Windows button on your keyboard and search for Julia. Press enter to open Julia

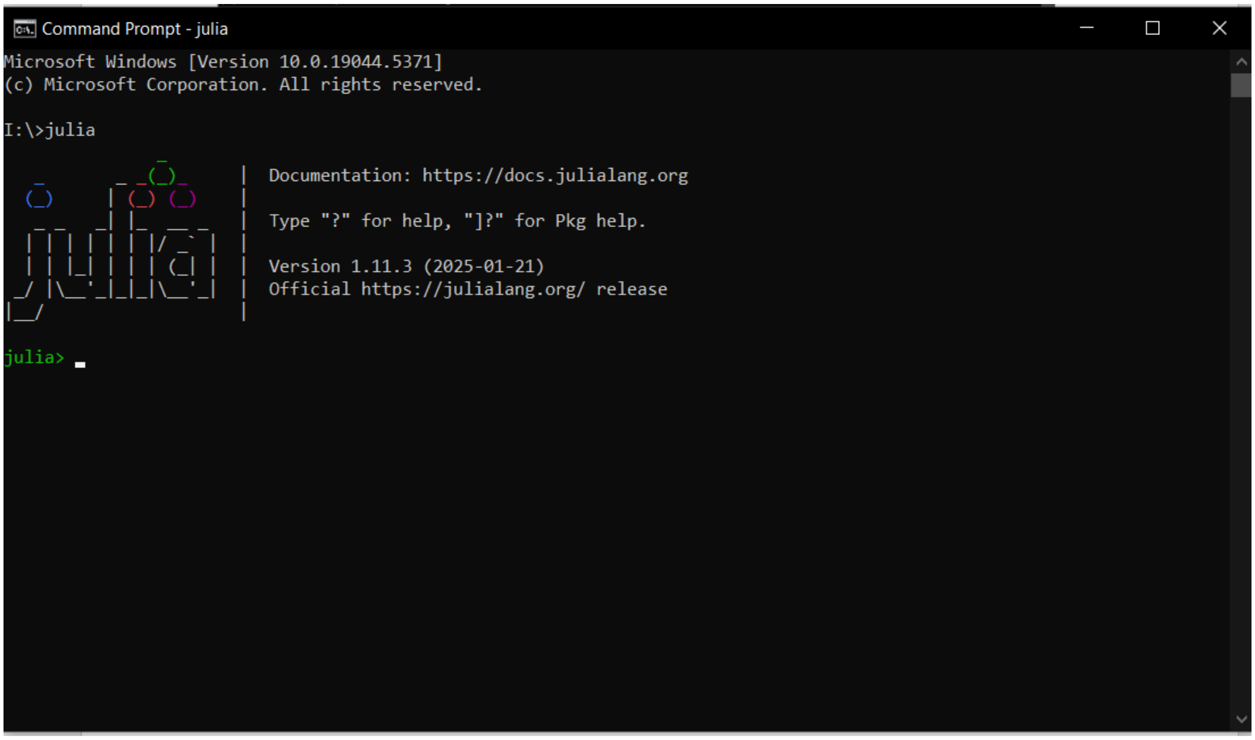




9. To double check whether Julia was correctly added to the PATH, open command prompt by pressing the Windows button on your keyboard and typing “CMD”. Press enter to run command prompt.



10. Type “julia” and press enter. The following should appear if Julia was added to the PATH (with the version being possibly different). If you get an error message, or do not see the following screen, maybe you forgot to add Julia to the path; return to step 1 and reinstall-.



11. You successfully installed Julia.

Installing Matlab and Dynare

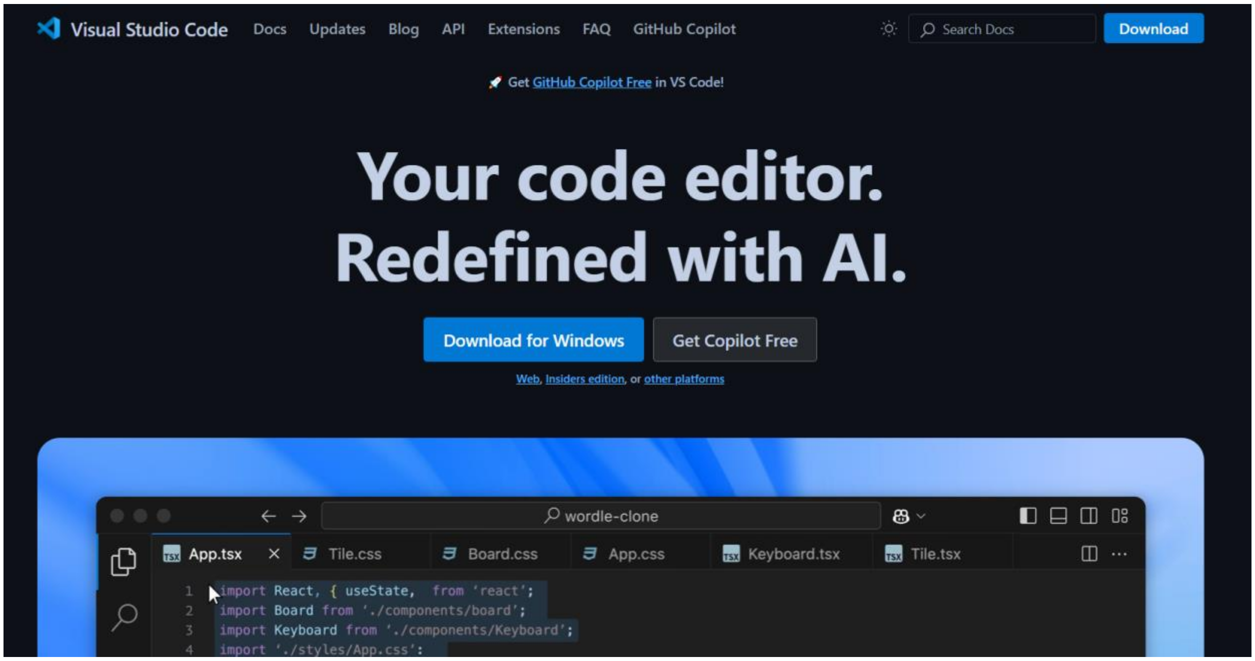
Parts of the replication material rely on Matlab and Dynare.

1. Go to https://www.mathworks.com/help/install/ug/install-products-with-internet-connection.html and follow the installation instructions for Matlab if you do not yet have Matlab installed. Alternatively, Octave can be used; however, we did not try this option.
2. To test whether Matlab was correctly installer, press the Windows button on your keyboard, search for Matlab and press ENTER. This should open Matlab.
3. Go to https://www.dynare.org/manual/installation-and-configuration.html and follow the installation instructions for Dynare. You can download the installer from this page https://www.dynare.org/release/windows/
4. To test whether Dynare was correctly installed, open Matlab and add the path to Dynare in the Command Window. Section 2.4 of the following link gives details of what the path would look like https://www.dynare.org/release/windows/. For example, for us the command looks like *addpath c:/dynare/5.5/matlab.* If no error is returned, everything should be working.

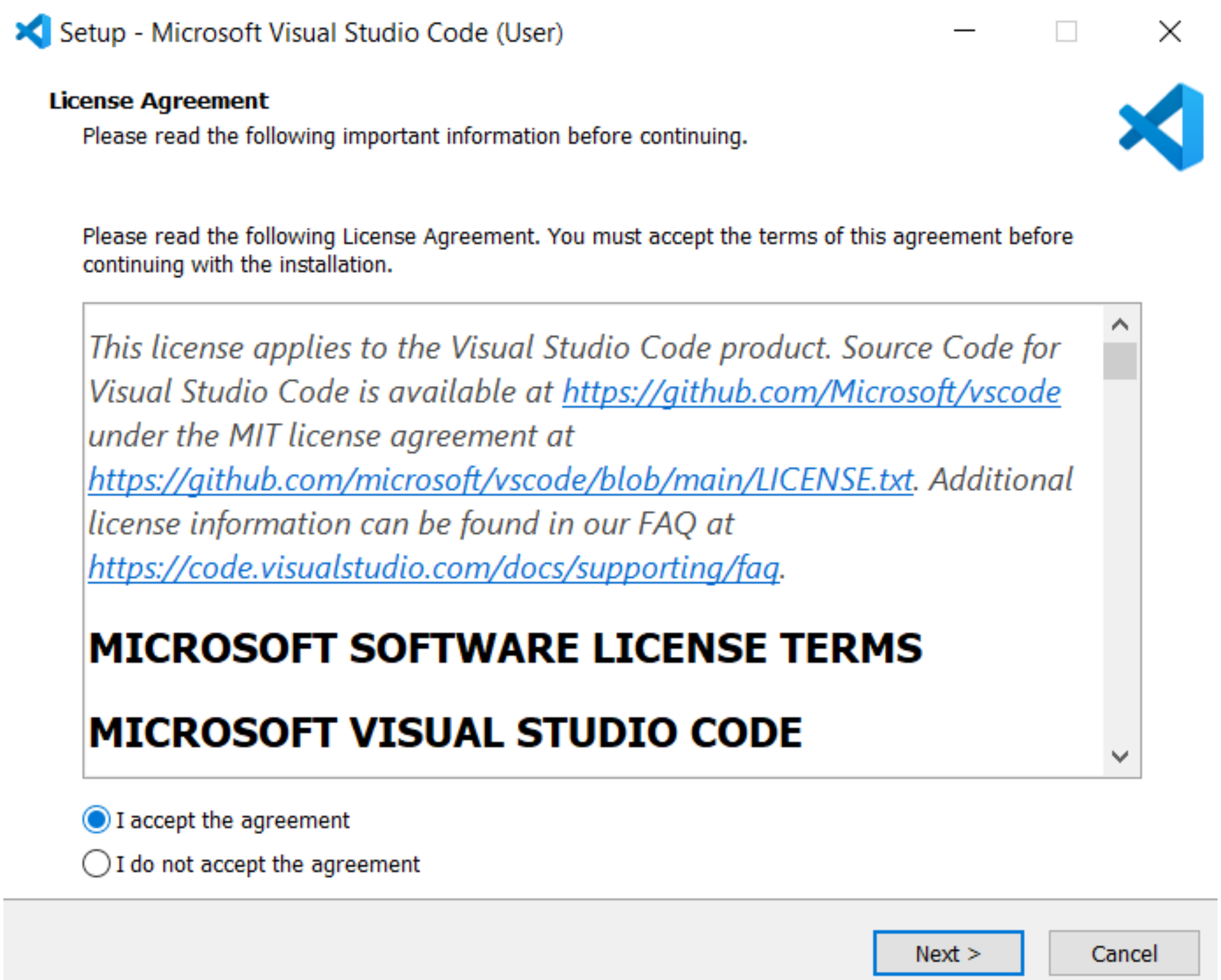
Installing VS Code

For all Julia related replication code, we recommend using VS Code as the development environment.

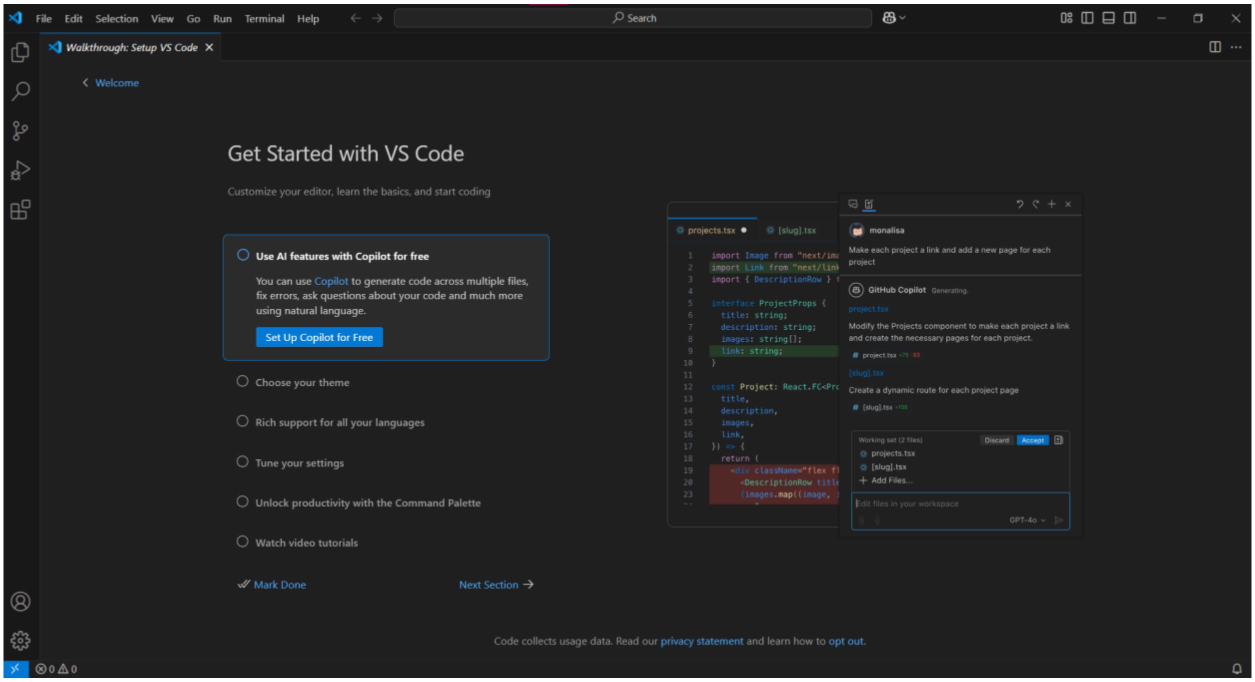
1. Go to https://code.visualstudio.com/ which should show you a website similar to the following.



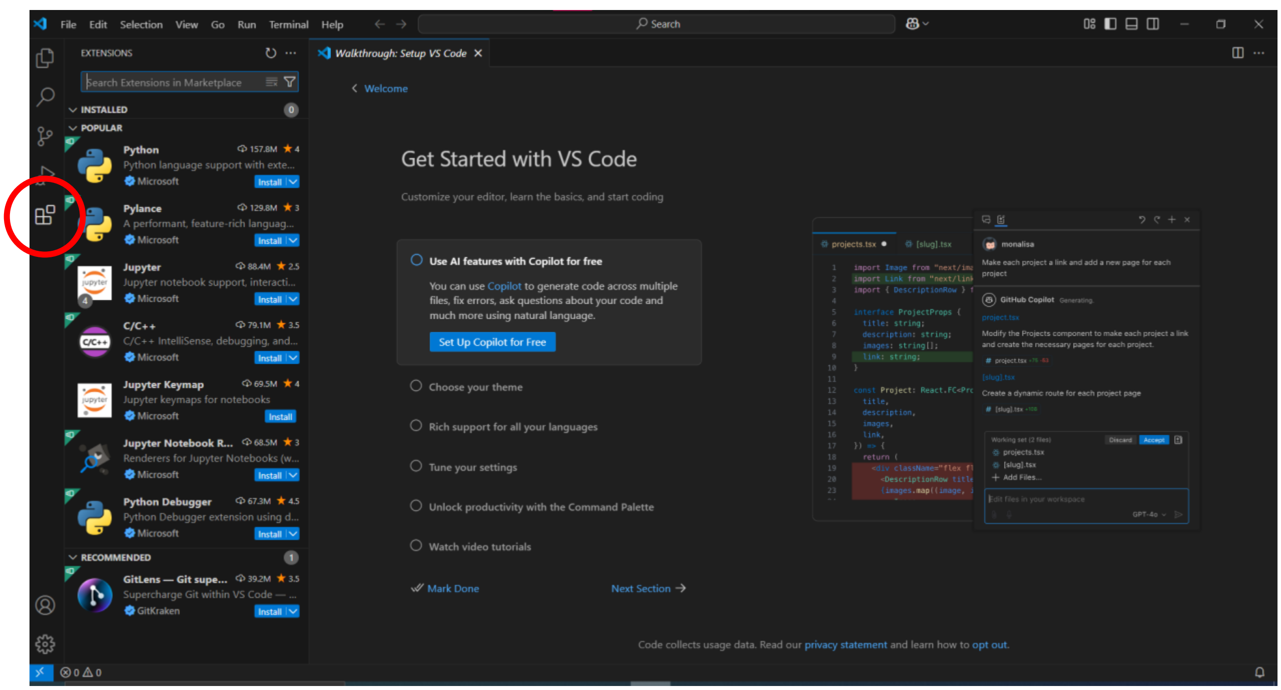
2. Press on “Download for Windows” to download the Windows installer.
3. The installer should now be in your Downloads folder. Double click the installer to open it.
4. Accept the T&Cs and press Next



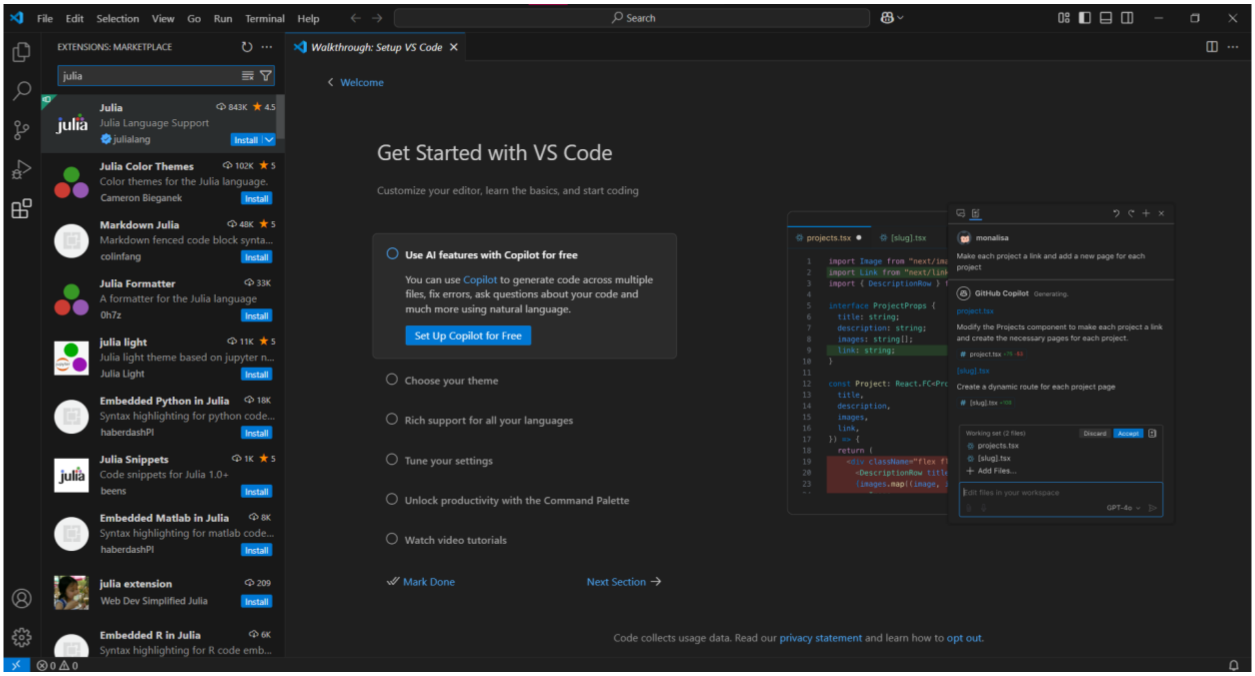
5. Press Next if you do not want to change the default location (most people should be fine with the default location). Otherwise, change the location by pressing Browse, and press Next after changing the location.
6. Press Next.
7. Press Next.
8. Press Install.
9. Press Finish to finish the installation and to run VS Code. This should open a window looking similar to the following.



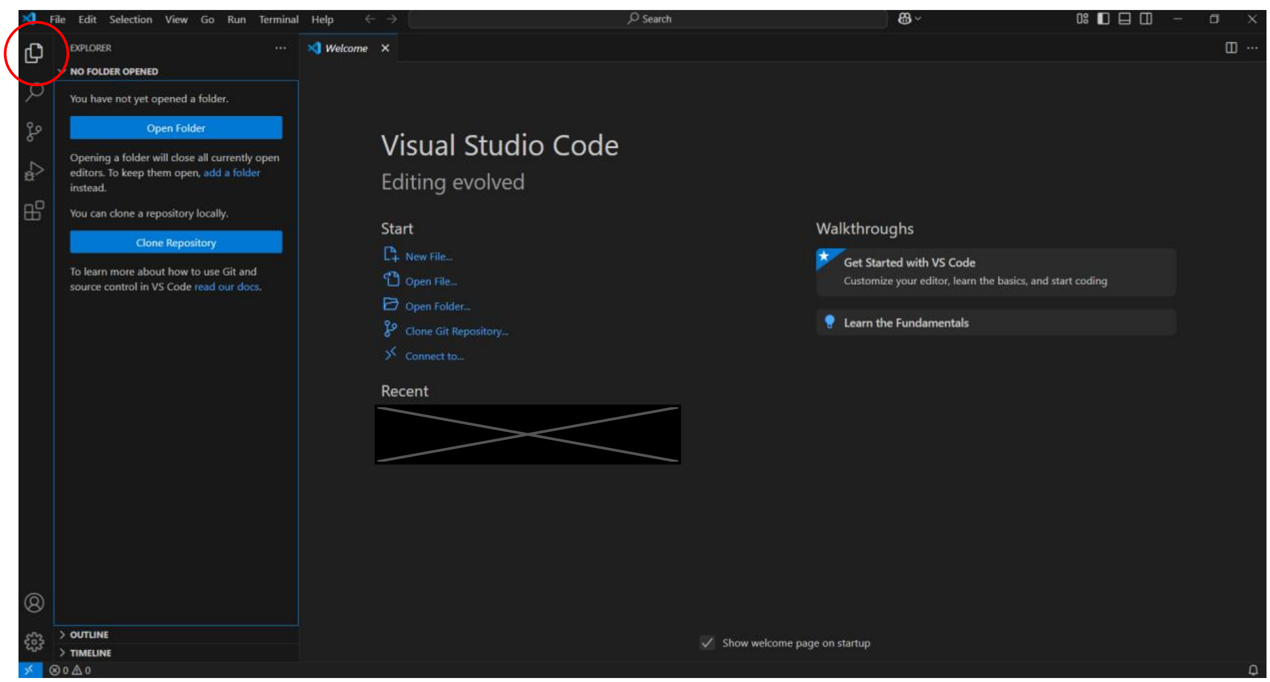
10. You can safely ignore all the “Get Started with VS Code” tasks. Instead, press the Extensions button on the left-hand-side. This is the 5th button from the top – the one consisting of four squares with one slightly offset from the others. This should open a sidebar and look similar to the following



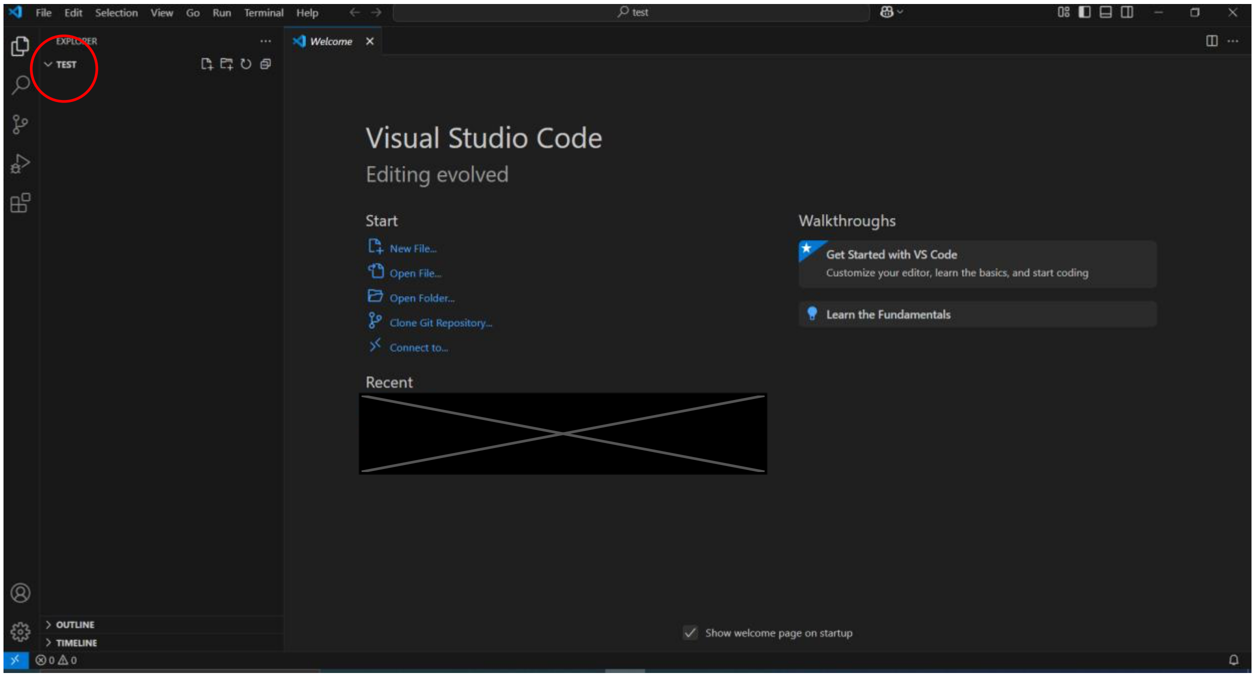
11. In the search box, search for “julia”. We are interested in the first search result, the “Julia Language Support”



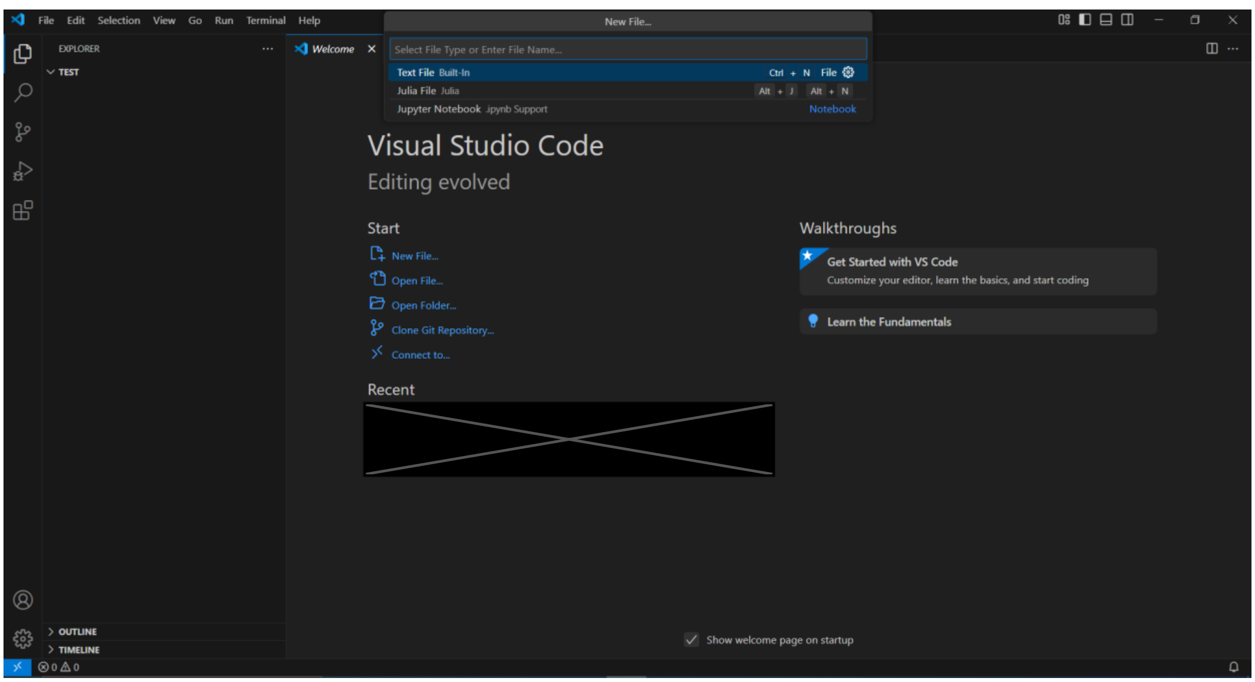
12. Press Install to install the extension. You might have to press “Trust Publisher and Install”.
13. To check whether the Julia Extension was correctly installed, press the explorer button, which is the first button from the top in the left menu.



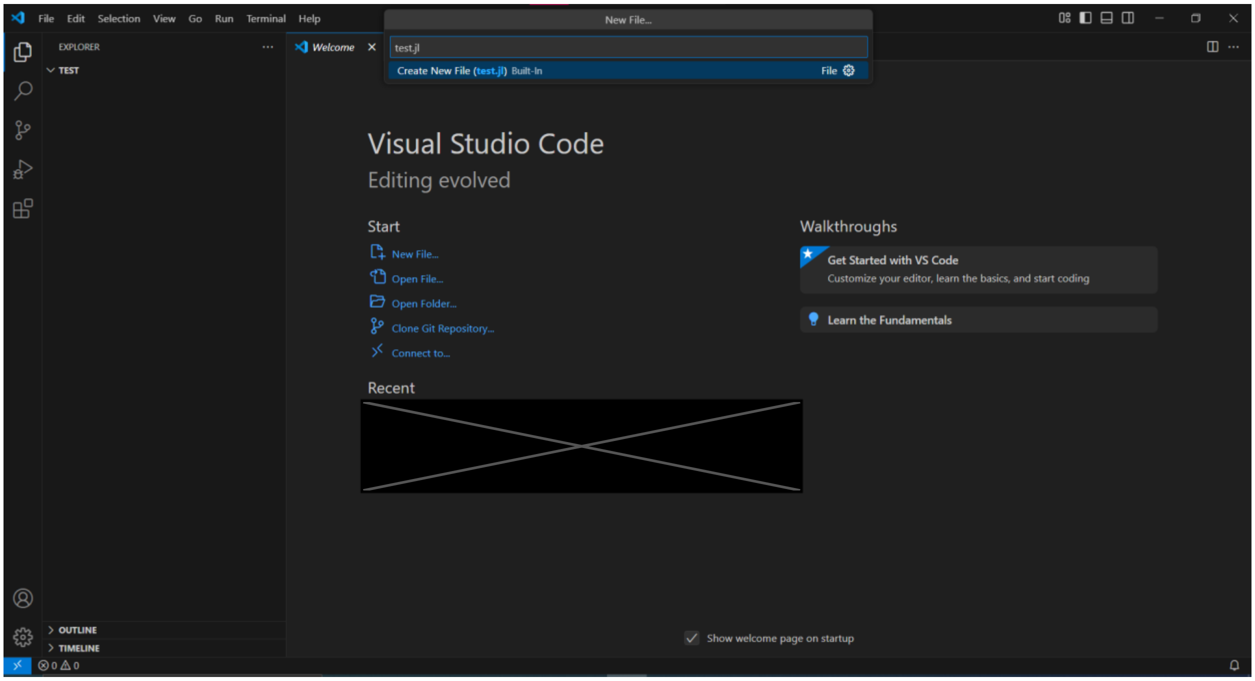
14. Press Open Folder and choose a folder in which we can create a Julia test file. This will open the folder in VS Code. The screen should look similar to the following, where “TEST” is replaced by the name of the folder you chose.



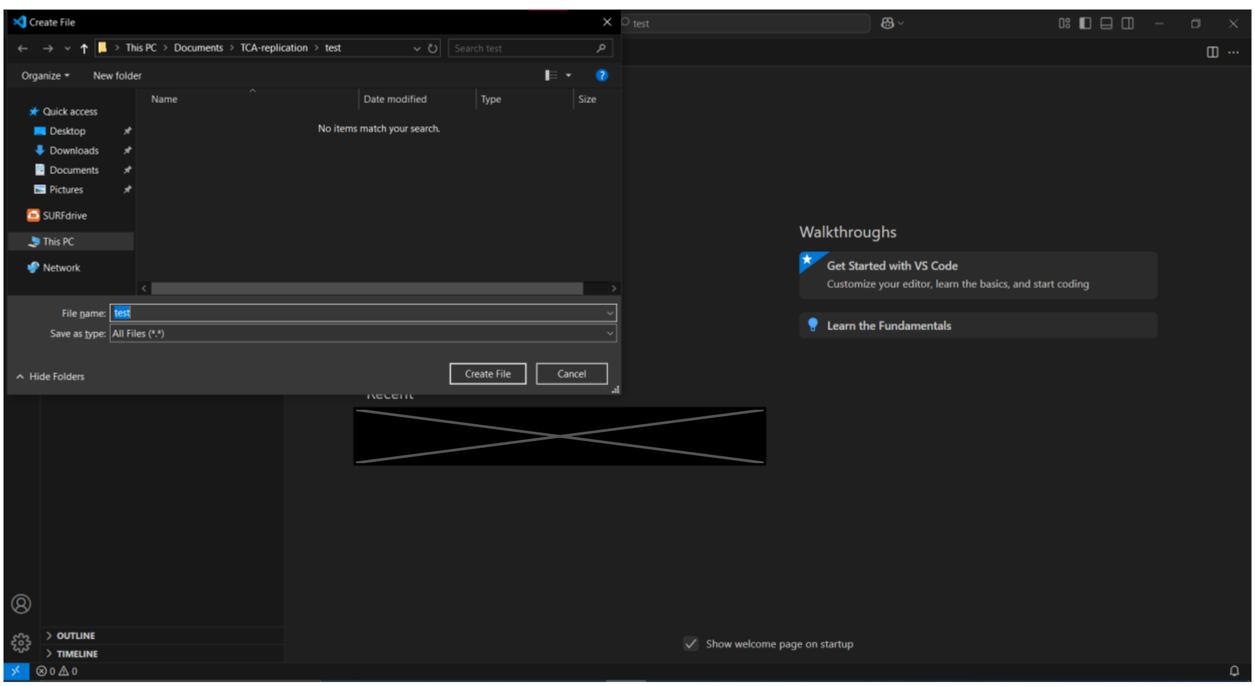
15. Go to File > New File. This should open a text box and your window should look something like the following.



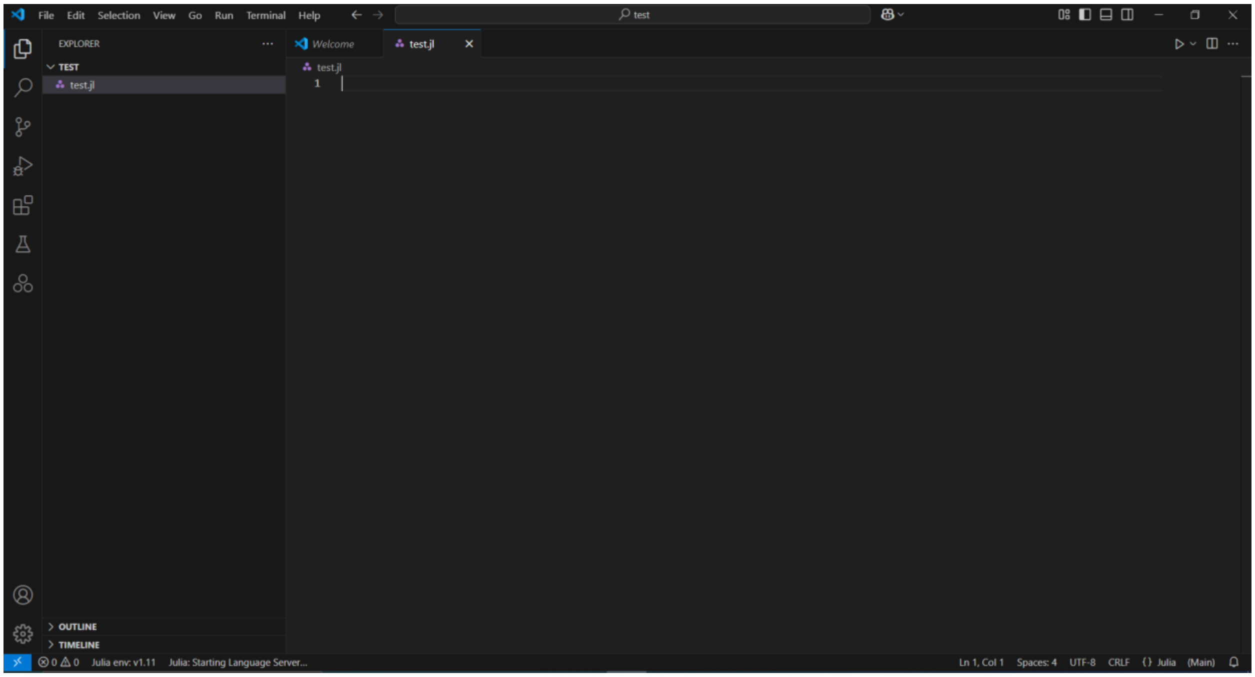
16. Type in “test.jl” and press ENTER.



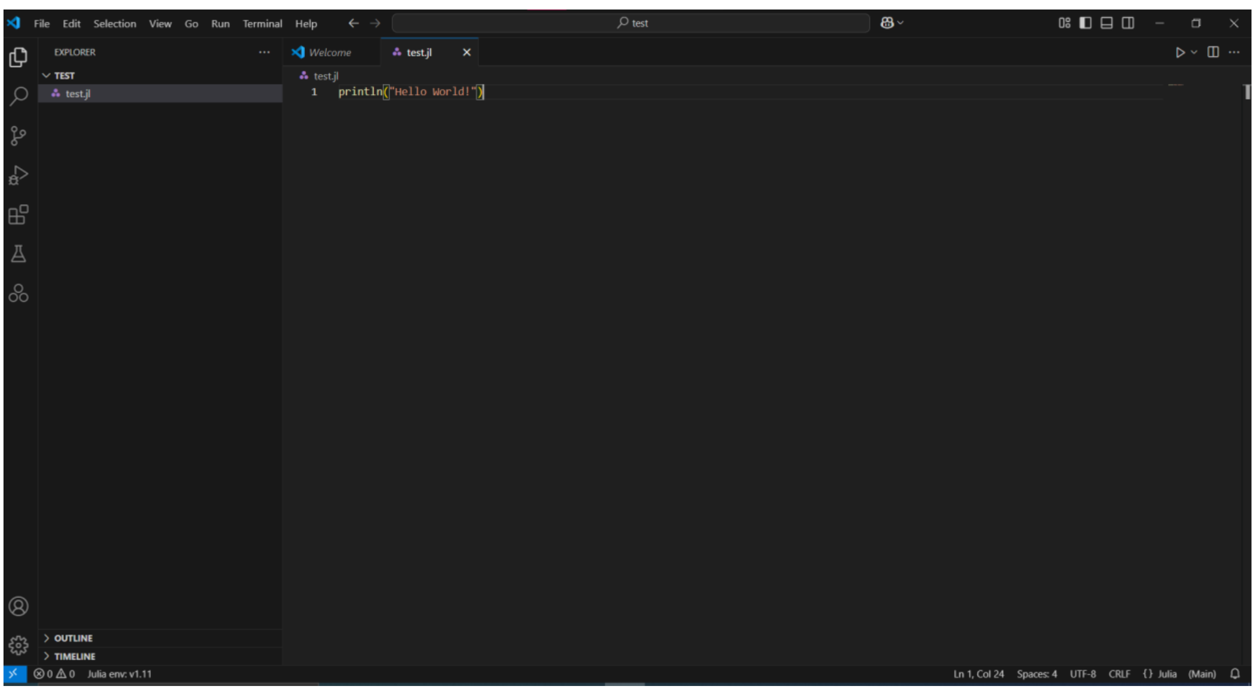
17. In the window that opens, you can simply press "Create File" to create the file in the folder you previously selected.



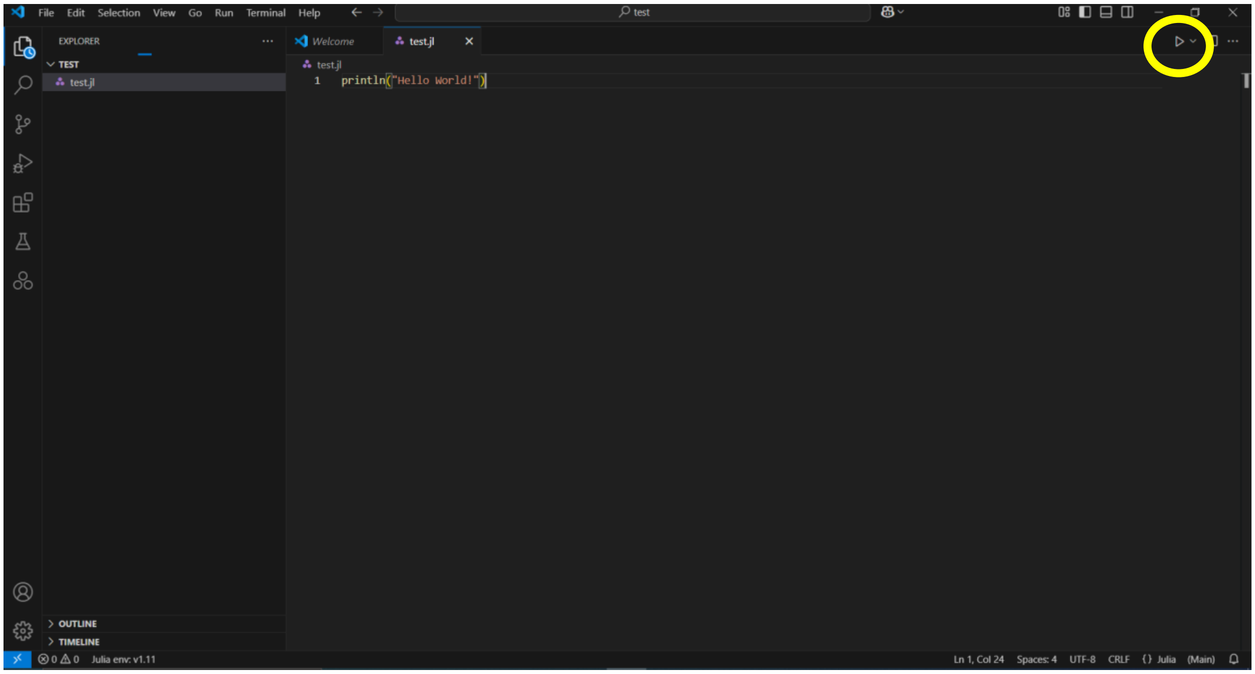
18. This should now open the "test.jl" file in VS Code. The window should look something like the following.



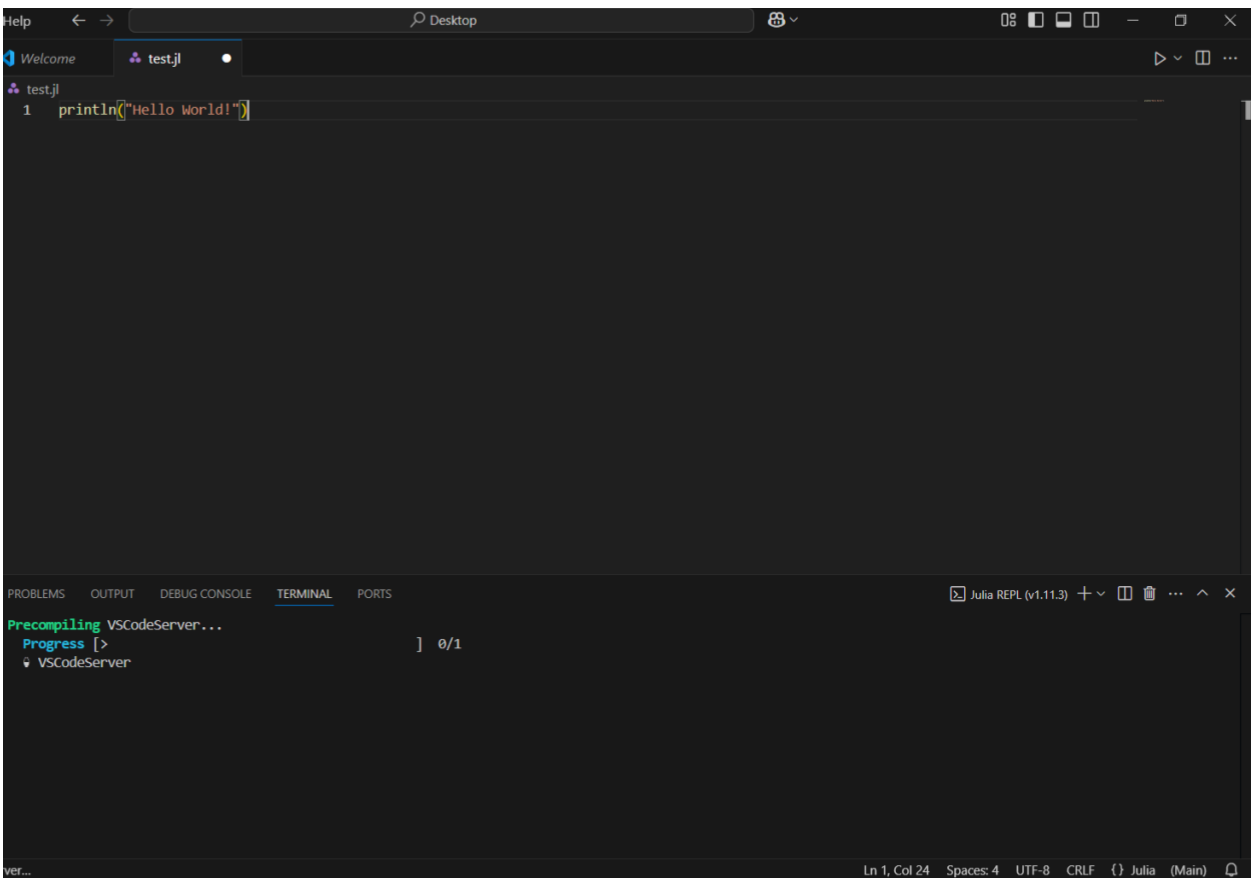
19. Type in `println(“Hello World”)`
    [Do not copy, please type it out].



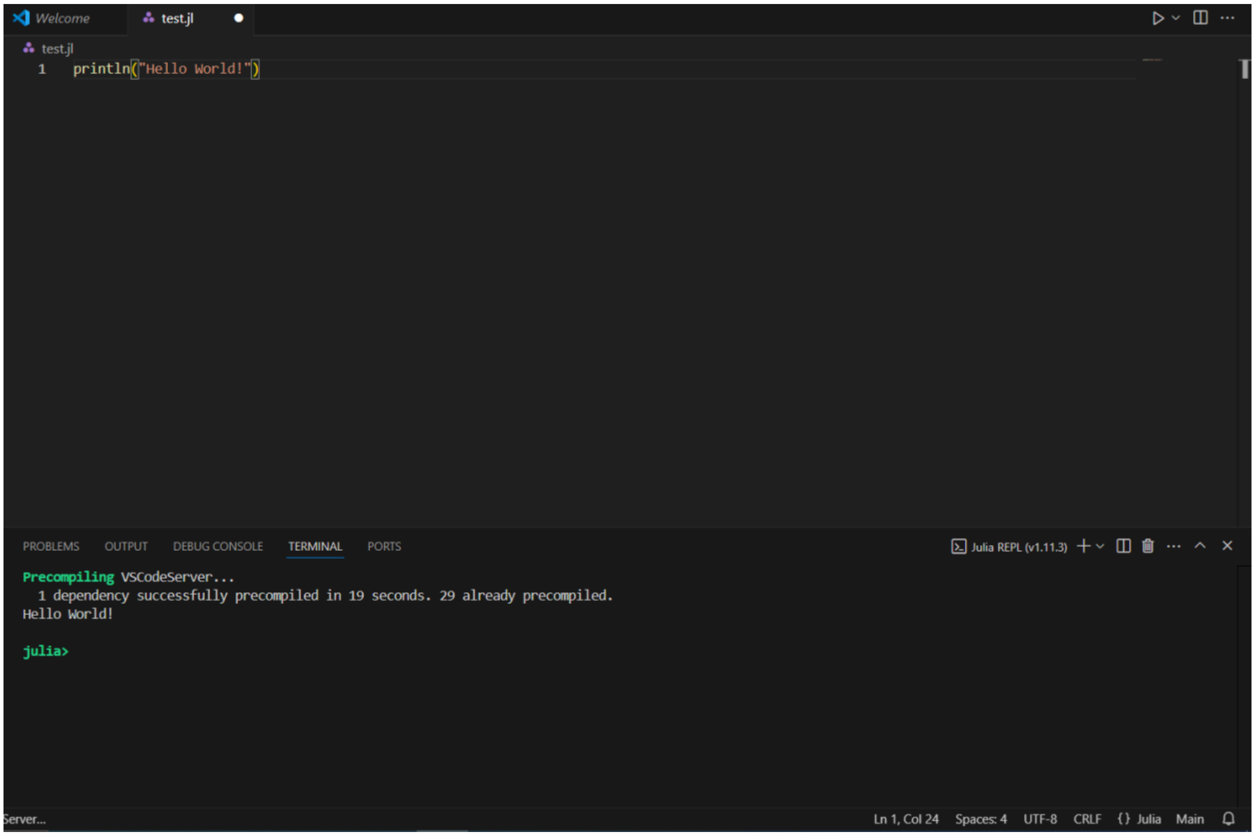
20. To run the Julia code, press the play button in the top right corner of the screen.



21. This will open a small Terminal window (usually on the bottom but it might be on the side). If this is the first time you run Julia from VS Code, you will see that it first installs the VSCodeServer. This might take a few minutes, but the code will run afterwards.



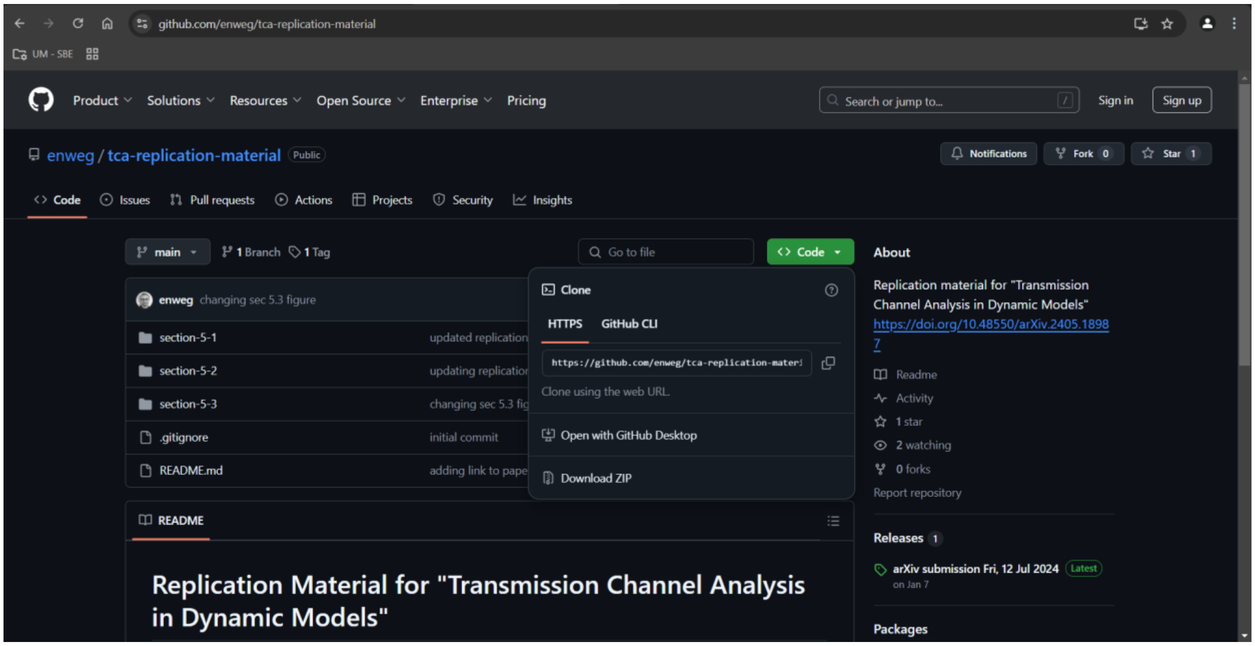
22. The output should finally look something like the following.



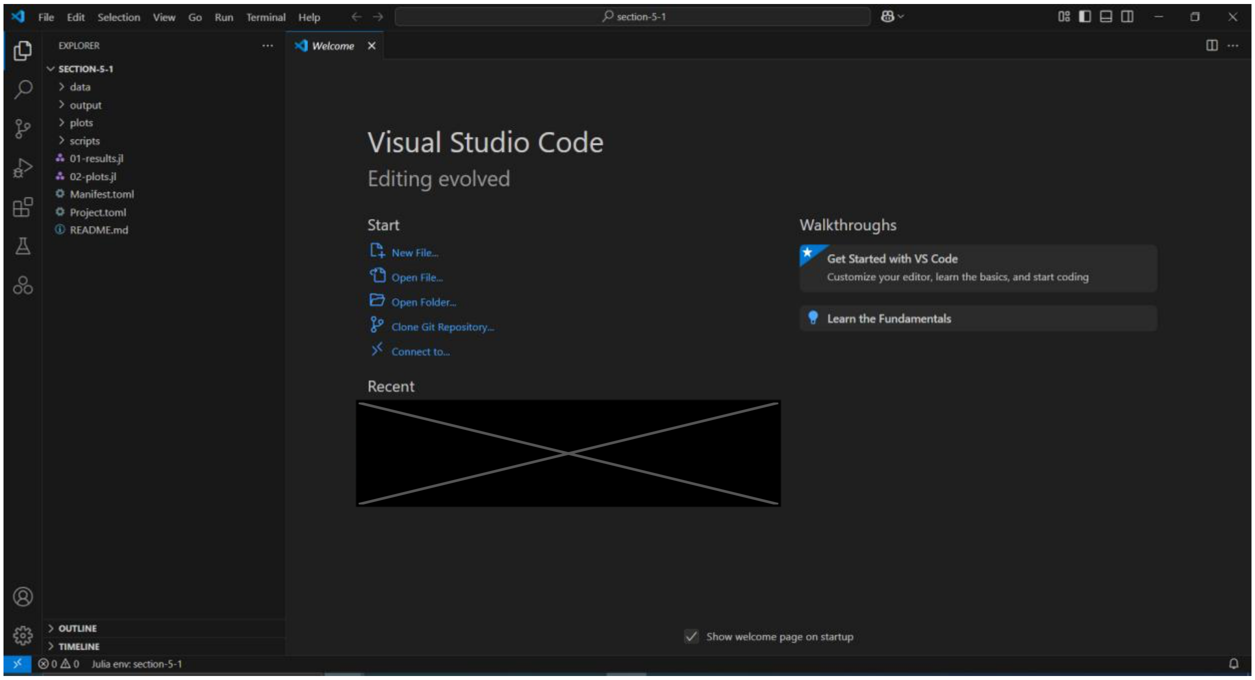
23. You have successfully installed VS Code and set up the Julia environment. You are now ready to replicate all Julia code

Replicating Section 5.1

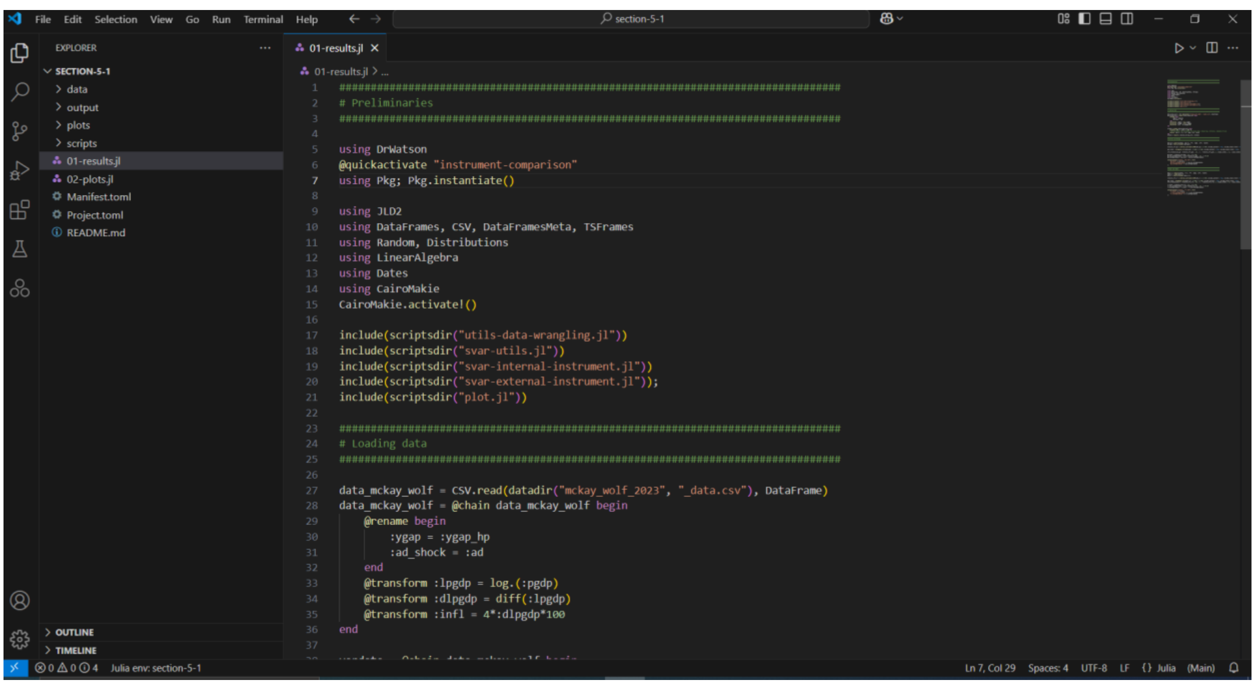
1. Go to https://github.com/enweg/tca-replication-material. There you should see a green button labelled "code". Press on the button and press "Download Zip".



2. Your Download folder should now contain the ZIP file containing the replication material. Move the ZIP file into a folder of your choosing.
3. Right-click on the ZIP file and press "extract all". Press "Extract" in the window that pops up.
4. You should now find a folder "tca-replication-material-main" in the folder you chose.
5. Go to VS Code. If you closed VS Code before, simply press the Windows button on your keyboard, search for VS Code, and press ENTER.
6. In VS Code, go to File > Open Folder. Browse to the replication material folder that you just extracted. Select the folder "section-5-1" and press Select Folder. Your VS Code should now look something like the following, with the folder contents listed on the left.

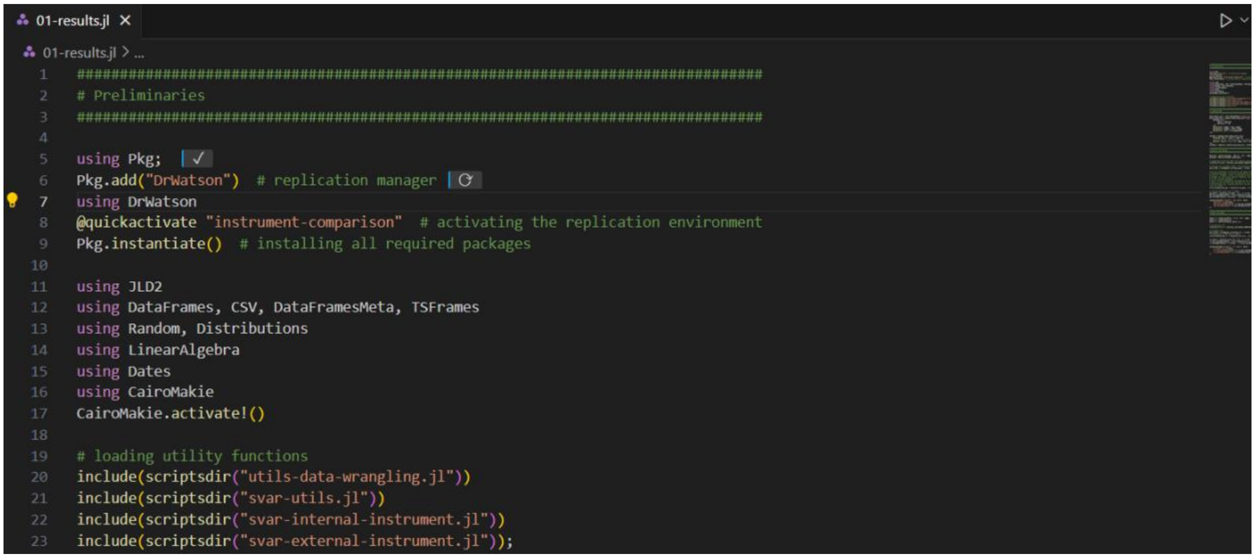


7. Open the file "01-results.jl" by double clicking on it. Your VS Code should now look something like the following



8. Two options exist to run the code. Either press the play button in the top right corner to run the entire file (same as before), or run the code line by line by putting the cursor in a line and pressing SHIFT + ENTER. A tick-mark will be shows behind lines that have been successfully run. A loading symbol will be shown behind lines that are currently being run. All outputs will be saved in the "output" subdirectory.

Many things need to be installed the first time you run it. This may take a while. We recommend taking a ~20min coffee break and checking back.

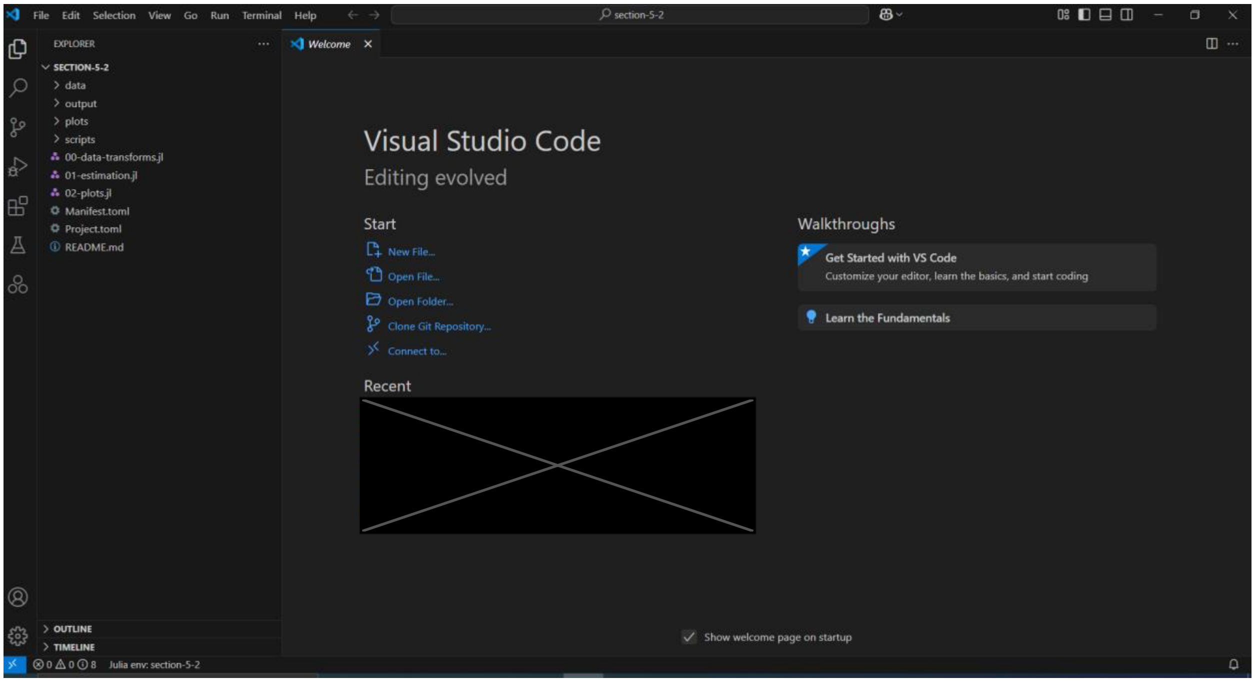



```julia
####################################################################################
# Preliminaries
####################################################################################

using Pkg;
Pkg.add("DrWatson")  # replication manager
using DrWatson
@quickactivate "instrument-comparison"  # activating the replication environment
Pkg.instantiate()  # installing all required packages

using JLD2
using DataFrames, CSV, DataFramesMeta, TSFrames
using Random, Distributions
using LinearAlgebra
using Dates
using CairoMakie
CairoMakie.activate!()

# loading utility functions
include(scriptsdir("utils-data-wrangling.jl"))
include(scriptsdir("svar-utils.jl"))
include(scriptsdir("svar-internal-instrument.jl"))
include(scriptsdir("svar-external-instrument.jl"));
```

9. To create the figures of Section 5.1, open the file "02-plots.jl" by double clicking it. Run the code either line-by-line by pressing SHIFT + ENTER, or all in one go by pressing the play button in the top right corner. All plots will be saved in the "plots" subdirectory.

Replicating Section 5.2

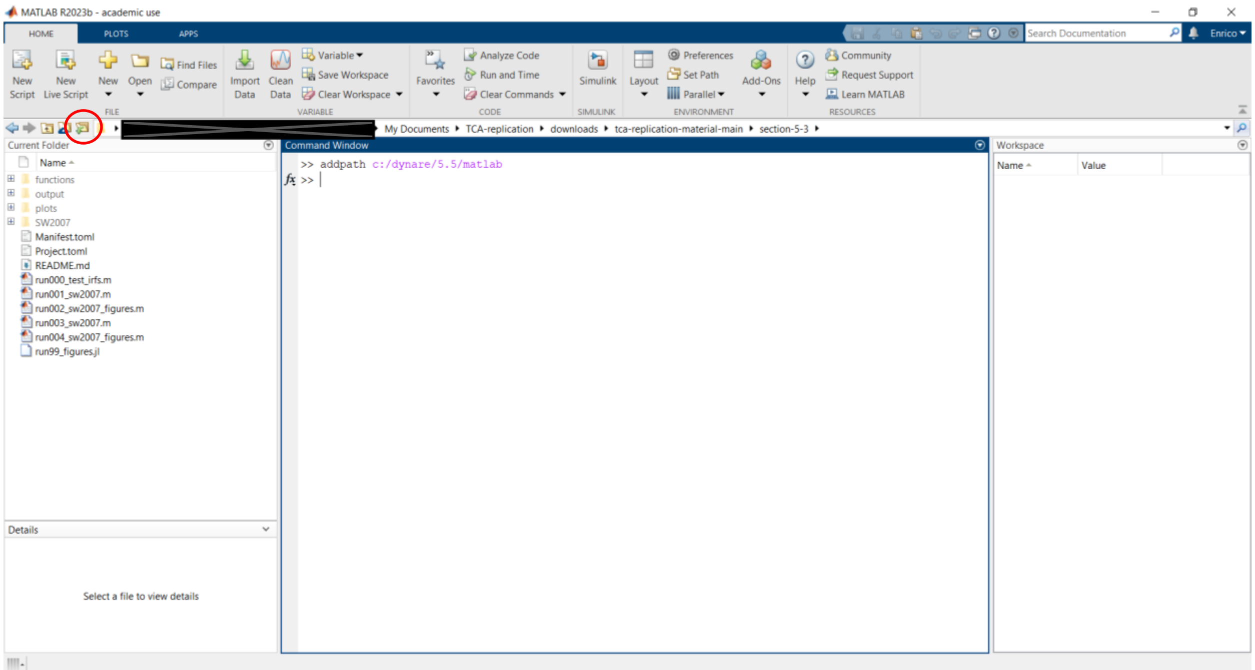
1. Follow steps 1 through 5 of section **Replicating Section 5.1**
2. In VS Code, go to File > Open Folder. Browse to the replication material folder that you just extracted. Select the folder "section-5-2" and press Select Folder. Your VS Code should now look something like the following, with the folder contents listed on the left.



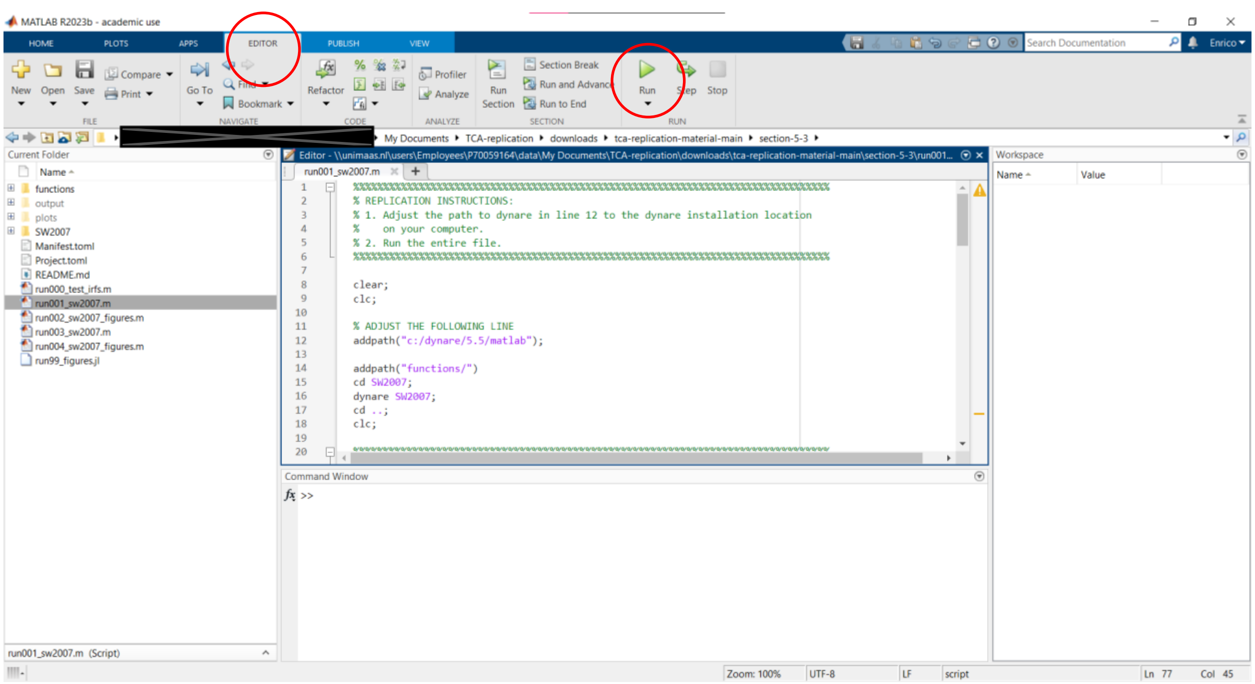
3. File "00-data-transforms.jl" creates the necessary datasets. It does not need to be run during replication. However, if you wish to run it, simply open it by double clicking it. Run the code either line by line using SHIFT + ENTER, or all in one go by pressing the play button in the top right corner.
4. To replicate the findings of Section 5.2, first open file "01-estimation.jl". Run the file either line by line using SHIFT + ENTER, or all in one go by pressing the play button in the top right corner. All results will be saved to the "output" subdirectory.
5. To replicate the plots of Section 5.2, open file "02-plots.jl" by double clicking it. Run the file either line by line using SHIFT + ENTER, or all in one go by pressing the play button in the top right corner. All plots will be saved in the "plots" subdirectory.

Replicating Section 5.3

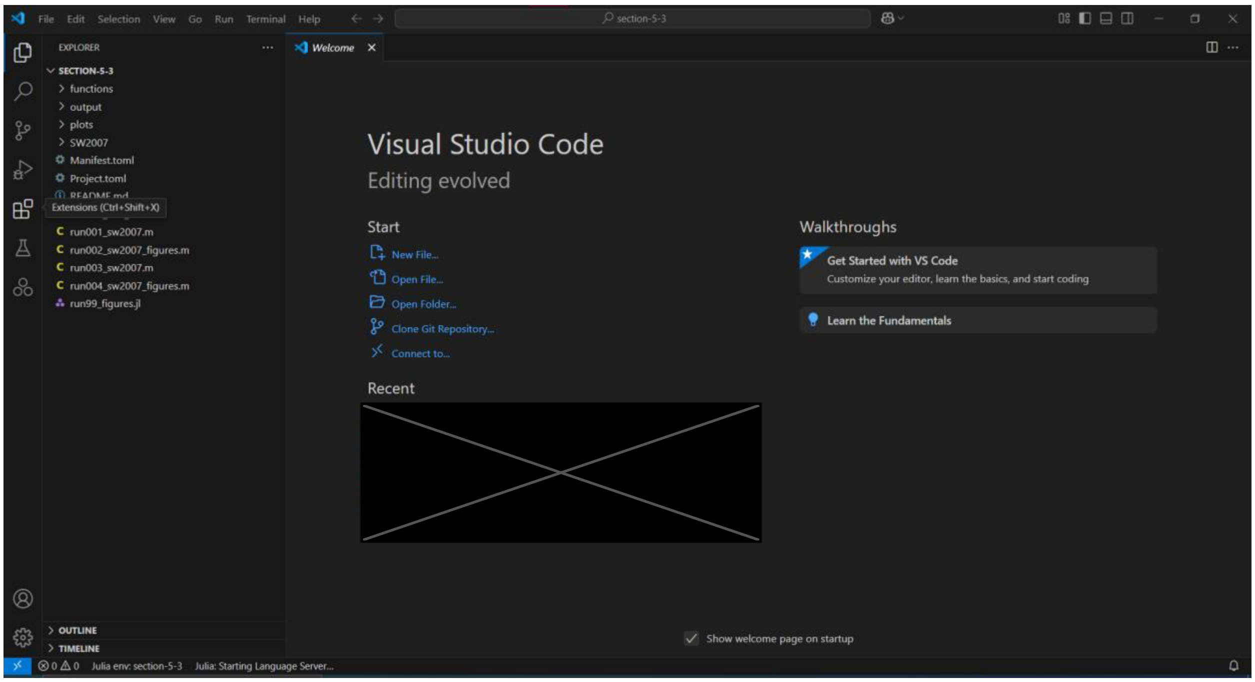
1. Follow steps 1 through 4 of section **Replicating Section 5.1**
2. Go to Matlab and browse the the "tca-replication-material-main/section-5-3" folder. This can be done by pressing the "Browse for Folder" button indicated below, browsing to the "tca-replication-material-main" folder extracted in step 1 and browsing into the "section-5-3" subfolder.



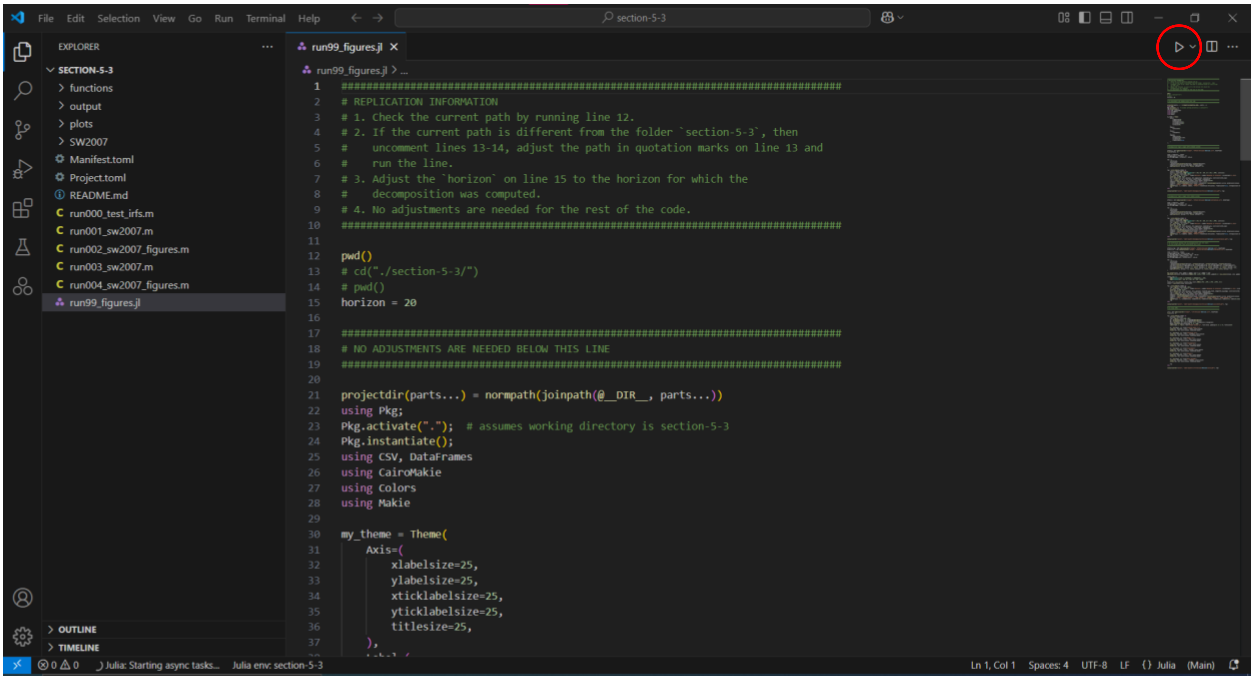
3. You should now see the contents of the folder in the left side-bar as indicated in the screenshot above. *Note: run00_test_irf.m does not need to be run in the replication. We will therefore skip those tests.*
4. Double click on run001_sw2007.m in the left side-bar to open the file in Matlab. To replicate the script, adjust the path to Dynare in line 12. This should be the same path as in the Section **Installing Dynare**. For example, for us, the path in quotation marks needs to be changed to "c:/dynare/5.5/matlab". To run the script, press the big green play button in the top menu bar (also indicated below). Make sure you are in the "Editor" tab (also indicated below). All results will be saved to the "output" folder.



5. To create Matlab figures (not used in the paper), follow the same steps for the files "run002_sw2007_figures.m", "run003_sw2007.m", and "run004_sw2007_figures.m". Matlab figures will be saved to the "plots" directory.
6. The actual figure used in the paper was created using Julia (for consistency reasons). To re-create this figure. Go the VS Code and open the "tca-replication-material-main/section-5-3" folder in VS Code. For more detailed instructions see Sections **Replicating Section 5.1** and **Replicating Section 5-2**.
7. After opening the folder in VS Code, the screen should look something like the following.



8. Double click on run99_figures.jl in the left side-bar to open the file. Press the play button in the top-right corner (indicated below) to run the entire file. The figure used in the paper, together with other figures, will be saved to the "plots" directory under the name "smets-wouters-decomposition-horizon-20-julia-both.pdf".



File Edit Selection View Go Run Terminal Help
section-5-3
EXPLORER
SECTION-5-3
functions
output
plots
SW2007
Manifest.toml
Project.toml
README.md
run000_test_irfs.m
run001_sw2007.m
run002_sw2007_figures.m
run003_sw2007.m
run004_sw2007_figures.m
run99_figures.jl
OUTLINE
TIMELINE
run99_figures.jl
# REPLICATION INFORMATION
# 1. Check the current path by running line 12.
# 2. If the current path is different from the folder `section-5-3`, then
#   uncomment lines 13-14, adjust the path in quotation marks on line 13 and
#   run the line.
# 3. Adjust the `horizon` on line 15 to the horizon for which the
#   decomposition was computed.
# 4. No adjustments are needed for the rest of the code.
pwd()
# cd("./section-5-3/")
# pwd()
horizon = 20
# NO ADJUSTMENTS ARE NEEDED BELOW THIS LINE
projectdir(parts...) = normpath(joinpath(@__DIR__, parts...))
using Pkg;
Pkg.activate("."); # assumes working directory is section-5-3
Pkg.instantiate();
using CSV, DataFrames
using CairoMakie
using Colors
using Makie
my_theme = Theme(
Axis=(
xlabelsize=25,
ylabelsize=25,
xticklabelsize=25,
yticklabelsize=25,
titlesize=25,
),
Julia: Starting async tasks...
Julia env: section-5-3
Ln 1, Col 1
Spaces: 4
UTF-8
LF
Julia
(Main)

macOS

Below, we provide a short overview of how to install Julia and VS Code on macOS. To replicate the results and to install Matlab and Dynare, follow the instructions outlined above.

Installing Julia

1. Go to https://julialang.org/downloads/ and download the .dmg file suitable for your version of macOS.
2. Double click on the .dmg file and drag the Julia app to Applications folder's shortcut.
3. To add Julia to PATH and to check if everything is running, open the macOS Terminal and run the following lines (below this corresponds to the installation of Julia 1.11, adjust the third line if you are installing a different version):

```
sudo mkdir -p /usr/local/bin
sudo rm -f /usr/local/bin/Julia
sudo ln -s /Applications/Julia-1.11.app/Contents/Resources/julia/bin/Julia /usr/local/bin/Julia
```

Installing VS Code

1. Navigate to https://code.visualstudio.com/ and click on the link “Download for macOS”. The download of the .zip file containing the VS Code app should start automatically.
2. Open (“unzip”) the .zip file and drag the VS code app into the Application folder.
3. Open VS Code by clicking on the app in the Application folder. (Since the app is not downloaded from the Apple’s App Store, access may be restricted. To allow macOS to open the app, adjust the settings under `System Preferences -> Security & Privacy`)
4. Follow the instructions above starting at 10.